Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/07/2007
PI Name:	Hulet, Randall G. Ph.D.		
Project Title:	Superfluid Phase Transition in an Ultracold Fermi Gas		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77251	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	5
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2006, per N. Hall-GRC (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2894		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity.		

Atoms have been observed to undergo Bose-Einstein condensation when the temperature is reduced to very low values. These atoms have integer spin and are bosons. An entirely different class of atoms, the fermions, do not directly exhibit this phase transition. However, fermions are predicted to form Cooper pairs, in analogy with the electronic pairing responsible for superconductivity, and at very low temperatures to form a superfluid. We are applying the technology used to produce Bose-Einstein condensation to achieve Cooper pairing in a gas of fermions, namely lithium-6 atoms.

The primary objective of this research is to detect a BCS-like phase transition in an ultracold gas of lithium-6 atoms. This type of phase transition requires that particles pair. Pairing is responsible for the phenomena of superconductivity in solids and superfluidity in liquid helium-3. It has never been observed in the gas phase, however, because densities are

Task Description:	usually too low for pairs to form and the temperatures to high for the manifestation of quantum statistical effects. In a previously funded NASA project, we showed that lithium-6 atoms exhibit an extraordinarily large attraction for each other, which is, in fact, larger than that of any known atom. Even for lithium-6, however, the critical temperature will be ~100 nano-Kelvin. In the past few years, the techniques of laser cooling combined with evaporative cooling in a magnetic trap have been demonstrated to be effective at reaching these low temperatures in gases of bosonic atoms, where they undergo Bose-Einstein condensation. Ultracold trapped atomic gases are remarkably versatile systems for investigations of many-body quantum effects, such as Bose-Einstein condensation. Unlike condensed matter systems, many of the key parameters, such as density, interaction strength, number, and temperature are under experimental control. This ability may enable us to explore the phenomena of Cooper pairing and fermion superfluidity under conditions never before achieved. Of particular significance, is the possibility of achieving Cooper pairing in the strong coupling regime, which has not been observed experimentally, nor has it been understood theoretically.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will lead to a greater understanding of the collective behavior of Fermi systems. This is especially important in the case of superconductors. Despite 20 years of intense effort, the exact mechanism of high-temperature superconductivity has yet to be discovered. We hope that the work that we are doing will provide insight into these important systems.
Task Progress:	This year was an especially productive one for our research on a superfluid Fermi gas of atomic 6Li. We have been exploring the phase transition that occurs when two different spin components of the 6Li atom undergo a pairing phase transition. The interactions between the atoms are controlled by a magnetically tunable Feshbach resonance. With the Feshbach resonance, it is possible to access the BCS-regime where the interactions are weak and attractive leading to correlated Cooper pairs, the BEC regime where the atom pairs can form weakly interacting bound molecules, and the strongly interacting regime between these two extremes. We published two papers on this system during the last grant year period. The first of these appeared in Physical Review Letters and contained three main results. First, we showed images of a pure molecular BEC. The fact that no thermal molecules were detectable demonstrated that the temperatures obtained in the experiment were significantly below the critical transition temperature. Secondly, by performing molecular spectroscopy of the pairs/molecules we gained an understanding of the microscopic nature of the pairs. The pairs are formed as a superposition of two free atoms and a bound diatomic molecular state. With the molecular spectroscopic technique we were able to measure the component fraction in the molecular channel. We found that this fraction varied from ~10% on the BEC side of the resonance to ~10-6 on the BCS side. Thirdly, the spectroscopic data can be related to the local pair correlation function, or equivalently, the superconducting order parameter. The measured order parameter shows excellent agreement for the expected dependence on interaction strength in both the BEC and BCS regimes, and smoothly interpolates in the strongly interacting crossover regime, where theory is not yet able to provide quantitative values. The second paper was recently published in Science. The topic of this publication was pairing when the numbers of the spin-up atoms differed from the number of spin-down. This difference causes a mismatch between the chemical potentials of the two spins and, according to usual BCS theory, should prevent ordinary pairing. We changed the relative polarization of the gas by driving rf spin-flip transitions between the two states. Our results are surprising and, I believe, remarkable. The fully paired superfluid is able accommodate the mismatch up to a critical polarization Pc, but for greater polarizations the gas undergoes a quantum phase transition to a phase separated state, where a fully paired superfluid core is surrounded by a shell of the excess unpaired atoms. We are currently doing experiments to understand the nature of the polarized, but non-phase separated superfluid for polarizations less than Pc. Supplemental Research Progress for Final Reporting We continue to study pairing in a polarized Fermi gas in the final year of the grant. We have made significant progress in understanding this complex system during this period. We find two superfluid phases, one in which the excess, unpaired atoms are excluded from a core of paired atoms, as described above, and a second, polarized phase, in which the excess atoms coexist with the paired superfluid. The phase separated state is favored at the lowest temperatures, while the polarized superfluid becomes stable at slightly higher temperature, and for relatively small number mismatch between spin-up and down atoms. In the phase separated state, the surface tension between the superfluid and normal states produces a marked deformation of the superfluid core. At small number mismatch, the core assumes the elongated "cigar" shape determined by the shape of the confining potential of the trap. As the number mismatch increases, the core becomes more oblate until at the highest polarization, its aspect ratio is 10 times smaller than that of the trap. This distortion enables us to distinguish the phase separated state from the polarized superfluid, for which there are no distortions. We are currently mapping out the phase diagram of the polarized Fermi gas as a function of temperature, interaction strength, and polarization. While we have currently identified three phases (two superfluid and the normal gas), there are predictions of exotic phases in which the pairs break spatial symmetries, which have yet to be convincingly observed in any system.
Bibliography Type:	Description: (Last Updated: 02/07/2007)
Articles in Peer-reviewed Journals	Partridge GB, Li W, Kamar RI, Liao YA, Hulet RG. "Pairing and phase separation in a polarized Fermi gas." Science. 2006 Jan 27;311(5760):503-5. Epub 2005 Dec 22. PMID: 16373534 , Jan-2006
Articles in Peer-reviewed Journals	Partridge GB, Strecker KE, Kamar RI, Jack MW, Hulet RG. "Molecular probe of pairing in the BEC-BCS crossover." Phys Rev Lett. 2005 Jul 8;95(2):020404. Epub 2005 Jul 7. PMID: 16090668 , Jul-2005
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Dissertations and Theses	Dries DF. "Fabrication of a High Resolution Relay Lens for Use in Imaging Ultra-Cold Quantum Gases." M.S. Thesis, Rice University, November 2005. , Nov-2005
Dissertations and Theses	Kamar RI. "Measurement of the Interactions in a Paired Zero Temperature 6Li Gas Throughout the BEC-BCS Crossover." M.S. Thesis, Rice University, January 2006. , Jan-2006
Papers from Meeting Proceedings	Hulet RG. "The Science of the Ultracold." Public Lecture at the Aspen Opera House, Aspen, CO, January 2006. Public Lecture at the Aspen Opera House, Aspen, CO, January 2006. , Jan-2006
Presentation	Hulet, R. G. "Pairing and Phase Separation in an Asymmetical Two-Component Fermi Gas" Physics Colloquium, University of Houston Jan-2006
Presentation	Hulet, R. G. "Pairing and Phase Separation in a Polarized Fermi Gas" Canadian Institute for Advanced Research Workshop on Ultracold Atoms, Banff, Canada Feb-2006
Presentation	Hulet, R. G. "Pairing and Phase Separation in a Polarized Fermi Gas" American Physical Society March Meeting, Baltimore, MD Mar-2006
Presentation	Hulet, R. G. "Pairing and Phase Separation in a Polarized Fermi Gas" Seminar at Quantum Information Research Institute, Los Alamos National Laboratory Mar-2006
Presentation	Hulet, R. G. "Matter Wave Soliton Train from a Bose-Einstein Condensate" Center for Non-Linear Science, Santa Fe May-2005
Presentation	Hulet, R. G. "Molecular Probe of Pairing in the BEC/BCS Crossover" American Physical Society Div. of Atomic, Molecular, and Optical Physics Annual Meeting, Lincoln, NE May-2005
Presentation	Hulet, R. G. "Molecular Probe of Pairing in the BEC/BCS Crossover" Workshop on Macroscopic Quantum Phenomena, Institute of Physics, Beijing Jun-2005
Presentation	Hulet, R. G. "The BEC/BCS Crossover in 6Li" Workshop on Macroscopic Quantum Phenomena, Nanjing University, Nanjing, China Jun-2005
Presentation	Hulet, R. G. "Molecular Probe of the BEC/BCS Crossover in 6Li" Workshop on New Developments in Quantum Gases, Institute for Nuclear Theory, University of Washington, Seattle Aug-2005
Presentation	Hulet, R. G. "Pairing in a Gas of Atomic Fermions" Physics Colloquium, University of Connecticut, Storrs Sep-2005
Presentation	Hulet, R. G. "Molecular Probe of Pairing in the BEC/BCS Crossover" Bose-Einstein Condensation: European Conference on Ultracold Gases and their Applications, San Feliu, Spain Sep-2005
Presentation	Hulet, R. G. "Bose-Einstein Condensation of Very Cold Atoms" World Year of Physics Symposium, Fermilab, Batavia, IL Oct-2005
Presentation	Hulet, R. G. "The BEC/BCS Crossover in an Atomic Gas" Physics Colloquium, Caltech, Pasadena, CA Oct-2005
Presentation	Hulet, R. G. "High Temperature Superfluidity at 50 nK" Princeton/Texas A&M Symposium on Bose-Einstein Condensation, Princeton University Oct-2005
Presentation	Hulet, R. G. "Einstein and Very Cold Atoms" Continuing Education Course, Rice University Oct-2005
Presentation	Hulet, R. G. "The BEC/BCS Crossover in an Atomic Gas" Physics Colloquium, Oklahomo State University, Stillwater, OK Oct-2005
Presentation	Hulet, R. G. "Pairing and Phase Separation in a Polarized Fermi Gas" Physics Colloquium, University of California, San Diego Jan-2006
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Significant Media Coverage	Cho A. "News of the Week: Condensed-matter physics. Mismatched cold atoms hint at a stellar new superfluid." Science. 2005 Dec 23;310(5756):1892. PMID: 16373549, Dec-2005